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REPORT

ON

JobInsight: An AI-Powered Job Market Analytics and Career Recommendation Platform

BY

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Certificate

This is to certify that, **Varad Bhandare, Vitthal Ijare** student of MSc Data Science and Big Data Analytics Semester III has successfully completed Mini Project in partial fulfillment of MSc Data Science and Big Data Analytics Sem-III under Dr. Vishwanath Karad MIT World Peace University, for the academic year 2026-2027.

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Place: Pune, Maharashtra

Date:

DECLARATION

This is to certify that this is the original work of **Varad Bhandare, Vitthal Ijare** completed under proper supervision. The work presented in this report has not been submitted anywhere against any course or program. The works are also unpublished and have not been submitted for publication anywhere. The ethics for preventing plagiarism are mentioned during the completion of this project.

Sign

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1. INTRODUCTION

Online job search platforms produce large amounts of job posting data, including job title, company name, location, experience requirements, required skills, and salary information. While this data is useful individually, it is difficult for job seekers to compare and interpret all of this data as a whole to understand the current state of the job market and their own suitability for various jobs.

JobInsight is an AI-powered job market analytics and career recommendation platform. It combines descriptive analytics, vocabulary-based skill extraction, a machine learning salary estimator, resume text extraction, TF-IDF-based job matching and skill gap identification into a single Streamlit interface.

The platform is designed to support career exploration rather than hiring decisions. The output of the platform is dependent on the quality and coverage of the available data and the limitations of the algorithms used.

1.1 Background and Motivation

Online job postings provide valuable insight into what skills are most in-demand, how different job families differ, what geographical patterns exist and what salary ranges are available. This information can be used by students and early-career candidates to determine which roles they should pursue and which skills they need to develop. [4], [5], [6], [12]

Job advertisements are not standardized; similar roles and skills may use different terminology, salary information may be missing and compensation may be listed in different currencies or pay periods. Therefore, before presenting reliable summaries, the platform must first clean and process the data. [6], [7], [8]

JobInsight links market-level analysis (e.g., common roles and skills) with candidate-level analysis (e.g., comparing resume content with job requirements). By linking both views into a single application, users can explore opportunities more systematically. [10], [11], [13]

The system is a decision support prototype and not an automated hiring tool. It cannot guarantee job suitability or salary accuracy and a skill not identified in a resume may still be possessed by the candidate.

The project utilizes data preparation, descriptive analytics, skill extraction, regression, document parsing, text similarity and skill comparison within a single application.

Job seekers typically need to know what roles are available, what skills are in demand, what salary patterns exist, what relevant listings exist and what possible skill gaps exist. JobInsight adds an analytical layer to traditional vacancy searches by analyzing a collection of postings and relating them to a candidate's resume. [4], [5], [10], [11]

1.2 Existing System

Traditional job search workflows focus on individual vacancies. Users search by title or location, open postings and manually compare requirements. Cross-posting analysis may require separate spreadsheets and can be time-consuming.

Market reports may display hiring trends without taking into account an individual's resume while resume tools may evaluate a document without demonstrating how its skills compare with the broader dataset. JobInsight seeks to connect these tasks while maintaining market summaries separate from candidate-specific similarity scores. [4], [5], [10], [11]

This comparison represents a common manual workflow and not every commercial job platform. The project focuses on integrating analytics and resume comparison for the selected dataset.

Common challenges include dispersed job information, skills embedded in unstructured text, inconsistent salary units, time-consuming resume comparisons and separation between market trends and individual career guidance.

1.3 Problem Statement

Job records contain useful but fragmented information. The system must standardize the data, identify skill signals, summarize available evidence and compare resumes with job descriptions while communicating the limitations of each result. [6], [7], [8], [10]

The primary tasks are to prepare job data, extract skills consistently, analyze market and salary patterns where comparable information exists and present understandable resume matching results without treating similarity as a guarantee of employment.

JobInsight addresses the need for one accessible workflow that combines job market summaries with resume comparison. It analyzes postings, identifies frequently mentioned skills, explores salary patterns, estimates salary from selected attributes, ranks job descriptions and highlights potential skill gaps. [4], [5], [10], [11], [13]

1.4 Objectives

Component-level evaluations can include checks for data consistency and duplicates, manual review of extracted skills and salary model metrics such as MAE, RMSE and R2 after verifying target units and validation design. Resume ranking should be evaluated with labeled relevance data if available; functional and usability tests can assess the application. Analyze job postings by role, company, location and available salary attributes. Extract recognized technical and professional skills from job descriptions. Calculate skill frequencies and summarize market demand. Train and use a regression pipeline to estimate salary from selected job features. Accept resumes in supported formats and extract their text. Rank job listings using TF-IDF and cosine similarity. Compare detected resume skills with job market skills to identify potential gaps.

Present results through an interactive Streamlit dashboard. Document limitations, evaluation needs and responsible use of candidate data.

1.5 Scope

The scope includes the provided job records, analytical summaries, the saved salary model pipeline, text extraction from supported resume formats, TF-IDF-based job ranking and identification of potential skills to explore.

The current scope does not establish live recruitment feeds, automatic applications, employer-side screening, production user accounts or real-time vacancy availability. The reviewed matching implementation uses TF-IDF and cosine similarity and not transformer-based semantic matching.

The intended users are students, job seekers and individuals exploring employment trends. The prototype supports analysis rather than recruitment management; extensions remain future work until implemented and tested.

The prototype uses CSV job data and a saved joblib salary model. The reviewed implementation does not establish live scraping, a production database, recruiter feedback or cloud deployment.

2. PROPOSED SYSTEM

2.1 System Overview

The system takes in pre-prepared job postings, verifies that all of the required information has been provided, and cleans up the data. Skill extraction looks at job postings and identifies skills based upon a defined list of words. Analytics provide an overview of the number of jobs available, the companies offering those jobs, where those jobs are located, the level of experience needed for each job, how much money you will earn doing each job, and the frequency of each skill being used in each job. [6], [7], [8], [14], [15] Salary estimation is done using a saved scikit-learn pipeline that uses a Random Forest regressor. Salary estimates are dependent upon both the labels used during training and the input features; however, you must ensure that your currency and pay-period are consistent prior to analyzing the results. [1], [2], [3], [15] Users can submit a resume (in a supported format) and the system will extract the readable text

from it. TF-IDF vectors and cosine similarity are used to rank the job postings, while skills identified in the resume can be compared against the skills listed in the dataset. All of this is displayed through Streamlit pages and visualizations. [10], [11], [13], [15], [16] Data files and model artifacts are shared among the different modules, therefore datasets, feature definitions, model files, and application code must all be compatible. If there are any changes made to the source data, then you may need to regenerate the processed files and models. A typical workflow would be: Market Exploration -> Resume Submission -> Matching List Review -> Identify Potential Skill Gaps. Results help support exploration; ultimately, users decide what path they want to take with respect to their career. In short, JobInsight is a single Streamlit application that combines prepared job data, skill extraction, market analytics, saved-model salary estimation, and TF-IDF resume matching.

2.2 Major Modules

Job Market Analytics: This summarizes the postings based on the available job attributes. The count is used to describe the data set and should not be used to estimate total labor-market vacancies. [4], [5], [6], [12] Skill Extraction and Intelligence: This identifies terms from a pre-defined vocabulary and sums up how many times they appear. The results are the number of times the term was identified, not all of the skills that an employer might need. [7], [8], [9] Salary Prediction: This loads the saved regression pipeline and uses the features that were expected. The estimates should include currency and pay period context if known; performance metrics should be reproducible before being described as validated. [1], [2], [3], [15] Resume Analyzer: This extracts text from supported documents. Scanned pages, tables and complex layouts may affect the quality of extraction, so incomplete or empty text should be addressed explicitly. Resume–Job Matcher: This uses TF-IDF and cosine similarity to rank job descriptions. It is a transparent baseline but may miss equivalent skills expressed with different wording. Scores indicate text similarity, not hiring probability. [10], [11], [13], [15] Skill Gap and Career Insight: This compares detected resume skills with skills mentioned in relevant jobs or the dataset. A missing detected skill is only a potential gap because it may be absent from the resume or missed by the extractor. Presentation: Streamlit provides navigation, filters, charts, document input and result tables. The interface should communicate empty states and limitations and avoid exposing unnecessary resume data.

Module	Main responsibility	Main output
Job Market Analytics	Summarize jobs, roles, companies, locations and salary attributes	KPIs, charts, filtered job data
Skill Extraction	Find known skills in job-description text	Skill lists and counts
Skill Intelligence	Aggregate extracted skills	Skill frequency and demand summaries
Salary Prediction	Apply saved Random Forest regression pipeline	Estimated salary for provided features
Resume Parser	Extract text from PDF, DOCX and TXT files where supported	Plain-text resume content
Resume–Job Matcher	Vectorize text and calculate cosine similarity	Ranked listings and similarity scores
Skill Gap Analysis	Compare detected candidate skills with job-demand signals	Potential missing-skill list
Dashboard	Expose analytics and tools through Streamlit	Interactive user interface

2.3 Operating Environment

The project will run locally using a python environment that has all of the required packages installed through requirements.txt. Virtual environments provide an additional layer of protection; however, since package versions are not fixed, it is recommended to test a "clean" install and record those version numbers so they may be reproduced. All data sets and models need to be placed in the locations where the application expects them. Using relative paths improves portability and also helps prevent issues related to available memory when working with csv files or creating text vectors.

Category	Technology / requirement	Purpose

Language	Python	Application logic and analysis
Data handling	Pandas, NumPy	CSV processing and numerical operations
Machine learning	scikit-learn	Preprocessing pipeline, Random Forest, TF-IDF and similarity
Dashboard	Streamlit	Interactive web interface
Visualization	Plotly	Interactive charts
Resume documents	PyPDF2, python-docx	PDF and DOCX text extraction
Model persistence	joblib	Load saved model pipeline
Input data	CSV files	Job postings and skill summaries
Recommended environment	Python environment with requirements installed	Reproducible execution

The supplied requirements include pandas, numpy, scikit-learn, streamlit, plotly, PyPDF2, python-docx and joblib. Package versions are not pinned and should be tested before submission.

2.4 Functional Requirements

The project will run locally using a python environment that has all of the required packages installed through requirements.txt. Virtual environments provide an additional layer of protection; however, since package versions are not fixed, it is recommended to test a "clean" install and record those version numbers so they may be reproduced. All data sets and models need to be placed in the locations where the application expects them. Using relative paths improves portability and also helps prevent issues related to available memory when working with csv files or creating text vectors.

Requirement	**Description**
Usability	Clear navigation, labels and readable charts.
Performance	Avoid unnecessary repeated processing of large CSV files and text corpora.

Reliability	Validate inputs and handle missing fields and unreadable documents.
Maintainability	Keep data processing, modeling and presentation code modular.
Privacy	Handle uploaded resumes carefully and avoid exposing personal information.
Reproducibility	Document data preparation, dependency versions and model evaluation.
Scalability	Recognize that local CSV loading and full-corpus text matching may become costly as data grows.

3. LITERATURE REVIEW AND RESEARCH GAP

The project synopsis categorizes related research into five areas: job-market analytics and skill gaps; skill extraction; salary prediction; resume-job matching and career recommendations. Researchers have applied various techniques including text mining, topic modeling, supervised learning, transformer-based representations, regression and recommendation techniques across all five areas. [1]-[13]

3.1 Job-Market Analytics and Skill Gaps

Research on job postings and education/course descriptions demonstrate how job-posting data can identify the skills employers are looking for and how those requirements may differ from what is offered in education. Topic modeling and other dimensionality-reduction techniques can help identify broad patterns in large collections of text. These types of studies provide a rationale for analyzing skill demand across different roles and locations. [4], [5], [6], [12]

3.2 Skill Extraction Methods

for extracting skills include everything from regular expressions and controlled vocabularies to supervised multi-label classification and transformer-based approaches. A predefined vocabulary is relatively easy to implement and interpret, and it is also cost-effective; however, it may fail to capture new skills, synonyms and contextual meanings. JobInsight utilizes a vocabulary-based baseline; however, semantic models could potentially be added as an improvement in the future rather than being a part of the current implementation. [7], [8], [9]

3.3 Salary Prediction Research

on predicting salaries typically utilizes job characteristics such as role, experience, education, industry and geographic location. Linear models can serve as a baseline; however, tree-based ensemble methods can capture non-linear relationships. Evaluating models requires careful consideration of salary units, currencies, missing values and possible duplicate/related postings between training and testing sets. [1], [2], [3]

3.4 Resume-Job Matching and Recommendation Matching

resumes can be accomplished using lexical similarity, semantic embeddings, supervised ranking or graph-based representations. TF-IDF and cosine similarity provide a clear baseline for comparing text documents; however, lexical similarity does not equate to job suitability or hiring probabilities. Ideally, evaluating recommendation quality should involve either labeled relevance judgments or ranking metrics. [10], [11], [13]

3.5 Research Gap

The following table lists the research areas reviewed, the approaches described in the literature, the gaps relevant to the proposed project and how JobInsight addresses those gaps. The table focuses on the integration gap identified in the project synopsis and does not imply that each technique is novel. The synopsis defines the research gap as an integration gap: the opportunity to integrate market-level skill and salary analysis with individual resume matching and skill-gap identification within a single workflow. JobInsight addresses this application-level goal by combining descriptive analytics, a salary regression model, text-based matching and candidate-oriented insights into a single dashboard. The project does not claim to have developed any of the individual algorithms. [4]-[13]

Sr. No.	Research Area	Existing Research / Approach	Identified Research Gap	How JobInsight Addresses the Gap
1	Job Market Analytics	Studies analyze job advertisements to identify employment trends and skill requirements	Market insights may remain separate from individual candidate analysis.	Provides job-market analytics through an integrated dashboard.

		using analytics and text mining.		
2	Skill Extraction	Research uses NLP, BERT-based models, supervised learning and predefined skill vocabularies.	Terminology variations, synonyms and different skill representations make consistent extraction challenging.	Uses vocabulary-based extraction to identify and summarize recognized skills.
3	Skill Demand Analysis	Studies analyze the frequency and distribution of skills across job advertisements and domains.	Candidates may find it difficult to connect market-wide skill demand with their own profiles.	Presents skill-frequency summaries and supports comparison with skills detected in resumes.
4	Salary Prediction	Studies apply regression, Random Forest and other ML methods to estimate salary from job attributes.	Salary data may be incomplete or difficult to compare across currencies, locations and pay periods.	Implements a Random Forest regression pipeline using available job features.
5	Resume–Job Matching	Research explores text similarity, neural networks and recommendation techniques to compare resumes	Matching may be separate from market analytics and may require recruitment interaction data.	Uses TF-IDF and cosine similarity to rank job listings against resume text.

		with job descriptions.		
6	Skill-Gap Analysis	Research examines differences between employer requirements and candidate or education skill profiles.	Market-level skill gaps may not be connected to an individual resume and relevant listings.	Identifies skills not detected in the resume that appear in relevant job requirements.
7	Career Recommendation	Recommendation research uses user profiles, job skills, preferences and market trends.	Career recommendations and job-market analysis are often studied as separate components.	Combines matching, skill-gap insights and market information to support career exploration.
8	Integrated Employment Intelligence	The reviewed literature addresses analytics, skill extraction, salary prediction, matching and recommendations through different methods and contexts.	The main gap is integrating market-level analysis with individual-level career intelligence.	Integrates these functions into one interactive Streamlit platform.

Research gap summary: JobInsight's main contribution will be the combination of market-level skill and salary analysis with resume-based job matching, potential skill-gap identification, and career recommendations. Rather than claiming to have developed a new algorithm, this project will apply existing methods within a single prototype.

3.6 Summary

The chosen methods will provide a working and interpretable prototype.

In addition to comparing the baseline skill extractor and TF-IDF matcher to other possible options, future evaluations may be conducted if there is labeled data and enough evaluation resources. [7]–[11]

4. METHODOLOGY AND IMPLEMENTATION

4.1 Data Collection and Preprocessing

The main data artifacts in this project's repository include jobs.csv, cleaned_jobs.csv, jobs_with_skills.csv and skill_frequency.csv. These raw and processed CSVs will be used to create a pipeline that takes raw job records and produces normalized data and summaries of skills. The preprocessing module will read the raw CSV; check for and remove duplicates; normalize the names of columns; trim text values; convert some of the numeric fields; calculate averages and ranges of salaries based on the minimum and maximum salary fields (if they exist); normalize some of the categorical fields; filter out records that do not have a title or job description; reset the index; and write cleaned_jobs.csv. [14], [15] • Duplicate removal removes exact duplicate records but does not necessarily remove near-duplicate job advertisements. • Missing numeric values are converted to null values if conversion fails. • Text fields are trimmed and selected fields are assigned placeholder values if specified. • Salary calculations are derived from the minimum and maximum salary fields that are available. Data-quality consideration: The code and synopsis show that salary currency and pay period can vary.
Therefore, a salary midpoint cannot be directly compared across different currencies or hourly, monthly and annual rates unless those values are standardized. This issue must be addressed prior to making cross-country salary comparisons or using salary predictions.

4.2 Skill Extraction and Market Analytics

The skill extraction module parses job description text to find specific skill names based on a pre-defined list; any identified skills are then saved for further analysis. The market analytics module processes the list of extracted skills, counts how many times each skill appears, and generates a summary of skill frequency. [7], [8], [9] This method produces transparent counts that can be used to create charts and tables.

However, the accuracy of this methods is dependent upon both the vocabulary and text matching rules. Employers often use synonyms, abbreviations, or new technology which could result in some skills being missed.

Additionally, many short terms are ambiguous without additional context.

Therefore, the skill counts produced through these rules should be viewed as mentions identified by the rules rather than an accurate representation of employer priorities.

4.3 Salary Prediction

The Salary Model Script will load the cleaned job data and check if there are all necessary fields (salary limits, job title, country and experience). Then it will create an average salary target based on the row-wise mean of salary_min and salary_max. After that, it will keep only those records which have a positive, non-null target. [1], [2], [3], [15]

The features used for modeling include title, country, years_experience_numeric, industry, function, experience_level, education_level, work_model.

However, only those features that exist in the dataset will be included in the model. Categorical features will be imputed and one-hot encoded while ignoring any unknown categories. Numeric experience will be imputed using the median. A RandomForestRegressor with 100 trees and a random state of 42 will be used as part of the pipeline.

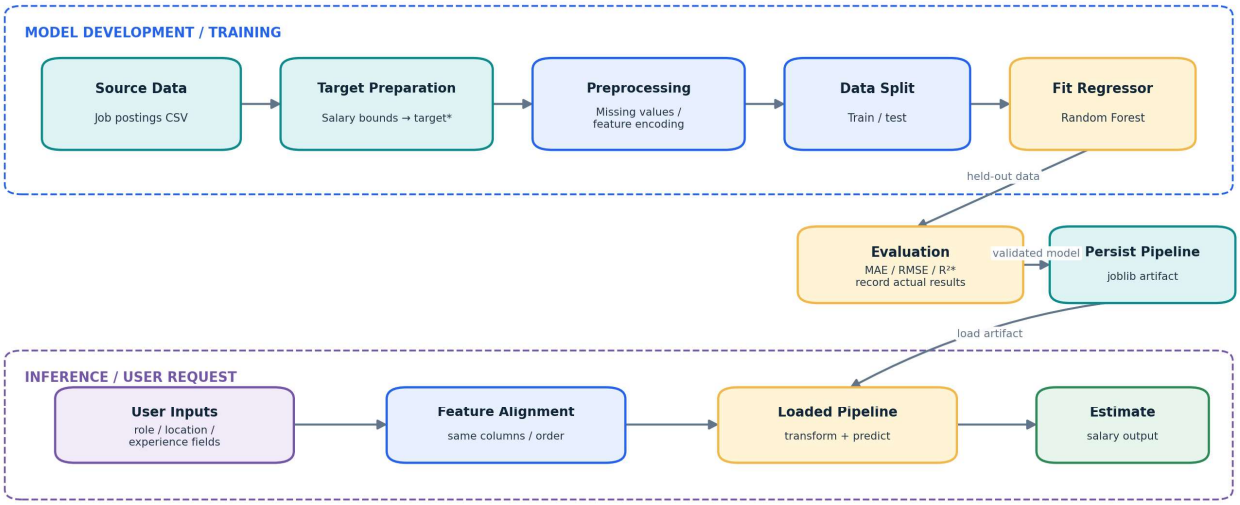


Figure 1: Salary Prediction Model Training and Inference Workflow

4.4 Resume Parsing

The resume parser will extract text from PDF, DOCX, and TXT files based on the project's documentation and dependencies. This extracted text will be used in the next step of the matching and skill-analysis process. The accuracy of the extracted text is dependent upon how well-structured the document is. Documents that have been scanned, contain only images, are formatted with multiple columns, or use unique fonts may result in incomplete or incorrectly organized text. At this time, OCR has not been included in the list of dependencies. [17], [18]

4.5 Resume–Job Matching

The matching module utilizes scikit-learn's TF-IDF vectorizer with English stop-word removal, unigram and bigram features, and a maximum feature vocabulary. Vectors for both resumes and job descriptions are compared via cosine similarity. Listings can then be ranked based upon how similar they are in terms of text to the resume. [15]

The score indicates lexical similarity within the chosen corpus. It should not be interpreted as the likelihood of being called for an interview or offered a position. The system could provide better explanations by showing which skills, job requirements and why a listing was ranked high. If the candidate-job search is limited to a subset of listings, this limitation should be stated.

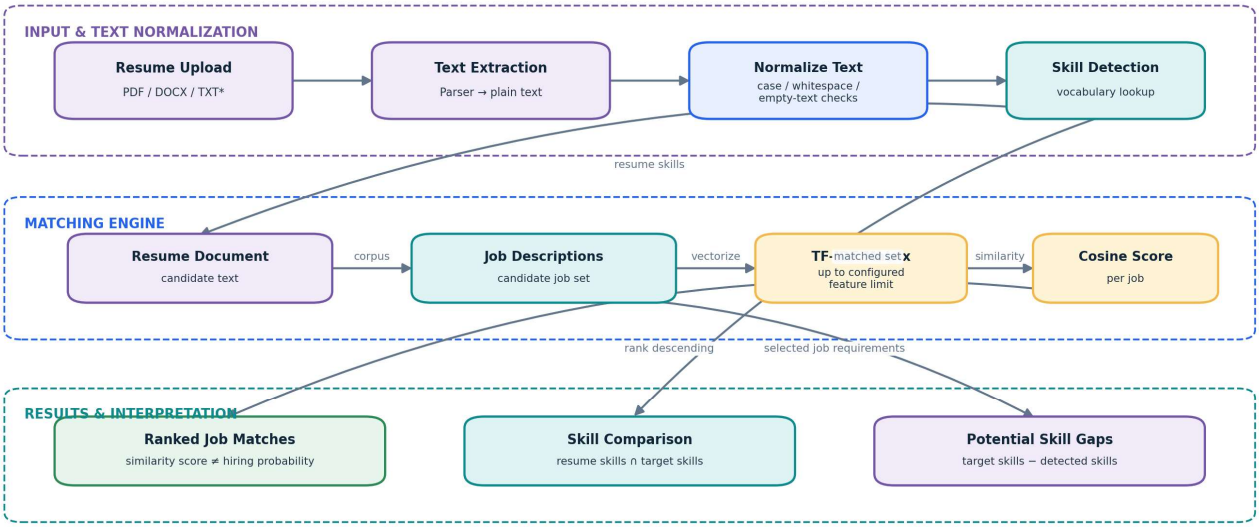


Figure 2: Resume Parsing, Job Matching, and Skill-Gap Analysis Workflow

4.6 Skill Gap Analysis and Career Insights

Skill-gap analysis compares skills recognized in a resume with skills associated with relevant jobs or frequent skills in the market dataset. The difference is presented as a list of potential skills to investigate or develop. A missing skill in extracted text does not prove that a candidate lacks that skill; it may simply be absent from the resume or missed by the extractor.

4.7 Integration

The Streamlit application acts as the presentation layer and combines market analytics, salary prediction, resume parsing, matching and skill-gap views. Data files and the saved model are loaded by the application or supporting modules. For reliable deployment, all paths should be project-relative, the prepared datasets should be versioned consistently, and expensive data operations should be cached or performed offline where appropriate. [16]

4.8 Data Quality, Validation and Reproducibility

The quality of the data is a critical first step as all subsequent components rely on the fields and text that remain after preprocessing. Prior to analyzing the data, the profile of the data set should include; the number of rows, the name of the columns, the type of

data within each column, the percentage of missing values, duplicate records, the number of categories in each categorical variable and the range of dates if there are posting dates. A data profile should accompany each generated data set so that changes in the source data can be identified rather than being misidentified as changes in the job market. Validation should differentiate between an empty field and a field that is truly empty.

For example, replacing a missing salary with zero would result in zero being analyzed as a valid salary observation by either summary statistics or a regression model. Likewise, a blank job description should not be treated as a description with no skills unless the record is specifically identified as having no usable text. All records that were removed during preprocessing should be accounted for and the reason for removal should be documented. The files jobs.csv, cleaned_jobs.csv, jobs_with_skills.csv and skill_frequency.csv should be viewed as related artifacts from a single data-processing run. If one file is updated and another is not, then the dashboard may display inconsistent counts or skill trends. A reproducible workflow should document the source file, processing date, script version and key configuration for each output. For a larger production system, a data version identifier or a manifest file could be used to link the processed files and saved model to the same release.

4.9 Salary Data Harmonization

Salary harmonization is treated separately because the same number may mean something completely different.

For example, 30,000 could be a yearly salary in one currency; a monthly salary in a different context; or a midpoint of a range in a different unit. To make accurate comparisons, you need to keep the original salary, currency, and rate unit; and then convert to a standardized value once you know the conversion rules and exchange rate date. Records that cannot be accurately converted should either be separated out or excluded from cross-currency comparisons, and not

simply combined. The target definition for model training must be clear. If average_salary is defined as the arithmetic mean of salary_min and salary_max, both salary_min and salary_max must refer to the same currency and pay period.

Additionally, the target should be reviewed for implausible values and inconsistent ranges (e.g., minimum is larger than maximum). If only one bound is provided, the project should indicate if the record is excluded or processed using a different rule.

Finally, the model report should include the target unit; the number of records retained; and any filtering criteria used.

4.10 Resume Data Handling and Privacy

Resumes can include name, email address, phone number, address, education history and employment history. The application should only extract and use the required information from the resume for the analysis being performed; it should never print the entire content of a resume into log files. Files uploaded by users should only be stored for the duration of their session unless they have been clearly told so and have provided consent. When deploying an application for multiple users, there are several key operational requirements including access control, cleaning up temporary files and providing a clear privacy notice.

Additionally, the system should inform users that any recommendations made are advisory. A missing skill within the extracted text of a resume could be due to how the document was laid out, a synonym that the vocabulary used by the application does not recognize, or a candidate's choice to exclude the skill from their resume.

Therefore, the application should use phrases like "not found in the provided resume" instead of saying "candidate lacks this skill." Using this phrase is more accurate and will help prevent making an unsupported judgment about a person.

4.11 Error Handling and Explainability

The errors can happen in many different ways: a CSV could have missing columns; a model file may be incompatible with the current library version; there may be no text to extract from a

resume; or a filter may return nothing. The application should provide clear messages when these types of errors occur and should never show raw stack traces or any type of sensitive user data on the interface. Detailed logs can be kept during development as long as they do not include resume content or other unnecessary personal information. One way to improve explainability is to show the evidence behind an output.

For example, a job match should show both the similarity score and a few examples of skills or terms that were found in both the resume and the job description. A skill-gap view should clearly differentiate between skills found in the resume and those commonly requested by the jobs you chose. A salary estimate should list all of the input variables used and explain that the estimate is based on historical training data and is not an offer from an employer nor a guarantee of what you will earn in the marketplace.

4.12 Reproducible Development Workflow

Only take screenshots once you have verified that what is being shown on screen matches your latest data and models. Open the dashboard and try all types of inputs (normal, empty, malformed, unsupported). Save both the model artifact and the data version/dependencies. Train the salary pipeline and save the evaluation metrics, target units, and split configuration. Create market summaries using the same version of the processed dataset. Extract skills and check a sample of the extracted skills against their corresponding job description. Preprocess the data and look at the number of rows generated, the type of each column, how many missing values exist, and if there are any duplicate rows. Keep a copy of the original dataset as it was created and note where it came from and how it can be used.

5. SYSTEM ANALYSIS AND DESIGN

5.1 System Architecture

The system can be represented as five logical layers: data, processing, AI and analytics, career intelligence, and presentation. The layers are a design view of the project; they do not imply that a separate production database or remote service is implemented.

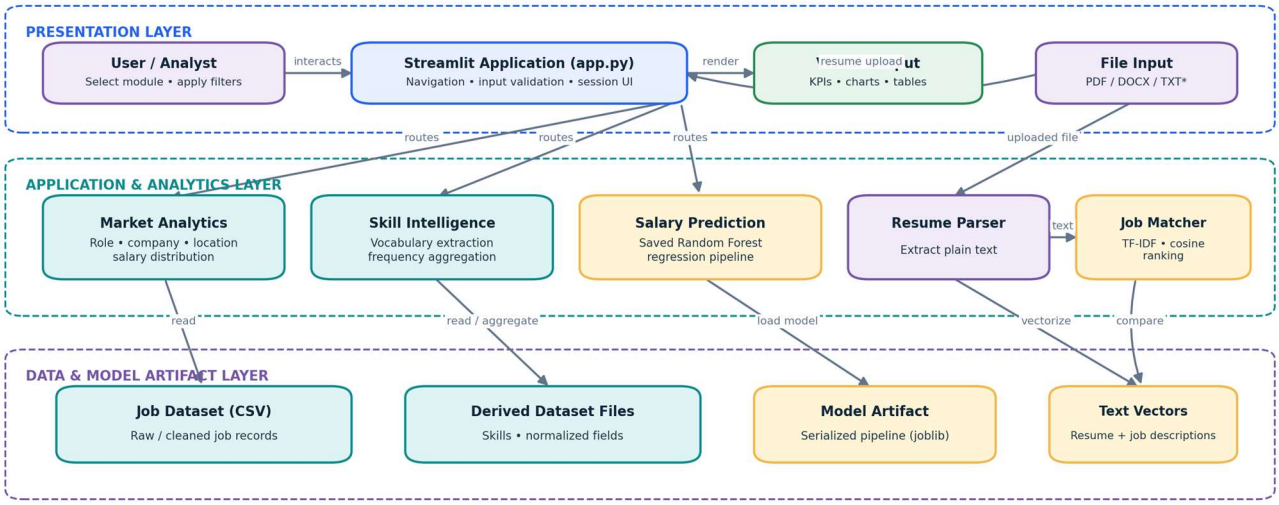


Figure 3: System Architecture of JobInsight

5.2 Workflow Diagram

The workflow starts with job data, performs validation and preprocessing, extracts skill signals, computes analytics and salary estimates, and optionally processes a candidate resume. The resulting matching and skill-gap insights are displayed in the dashboard.

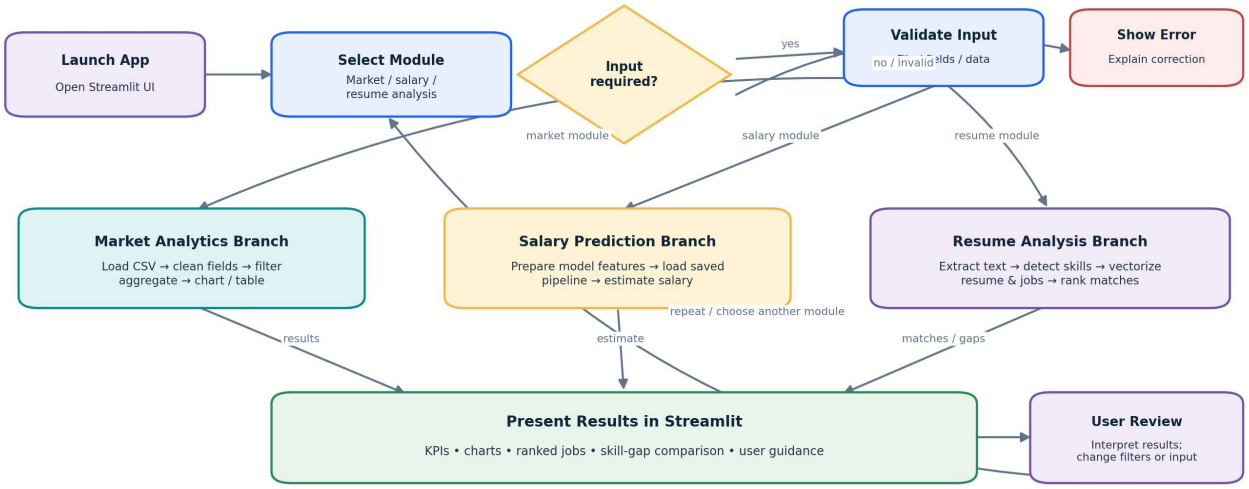


Figure 4: JobInsight Application Workflow Diagram

5.3 Data Flow Diagram — Level 0

At a high level, the system accepts job-posting datasets and resume input. The processing components generate market summaries, salary estimates, ranked job listings and potential skill gaps. The user interacts with the outputs through the dashboard.

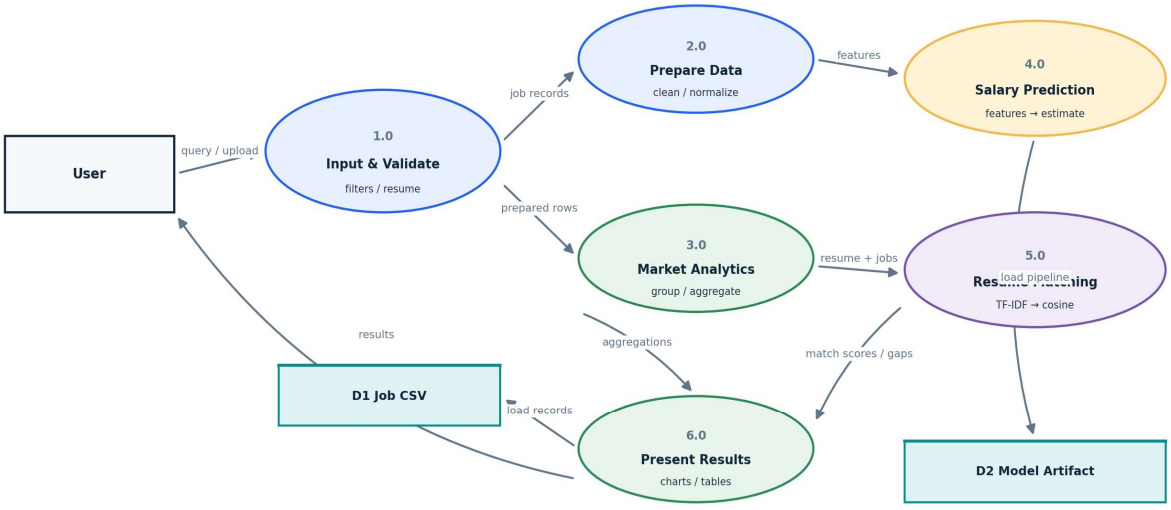


Figure 5: Jobinsights Data Flow Diagram (DFD- Level 1)

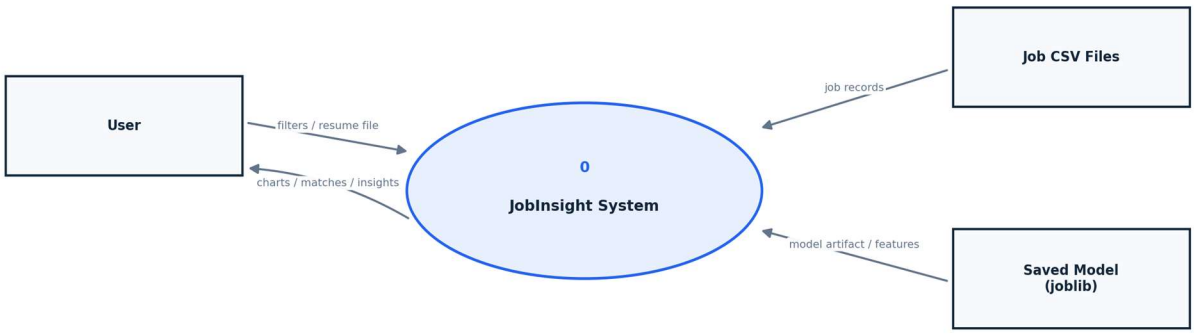


Figure 6: JobInsight System Context Diagram (DFD- Level 0)

5.4 Use Cases

Actor	Use case	Description
Job seeker	Explore job market	Review role, company, location and salary summaries.
Job seeker	Analyze skills	Inspect frequently detected skills and demand patterns.
Job seeker	Upload resume	Provide a supported resume file for text extraction.
Job seeker	Match jobs	Receive ranked listings based on text similarity.
Job seeker	Review skill gaps	View skills detected in the resume and potential gaps.

Job seeker	Estimate salary	Enter available job attributes for an estimated salary output.
Administrator / maintainer	Prepare data	Run preprocessing and skill analytics scripts.
Administrator / maintainer	Maintain model	Train, evaluate and replace the saved salary pipeline.

5.5 Data Dictionary — Core Fields

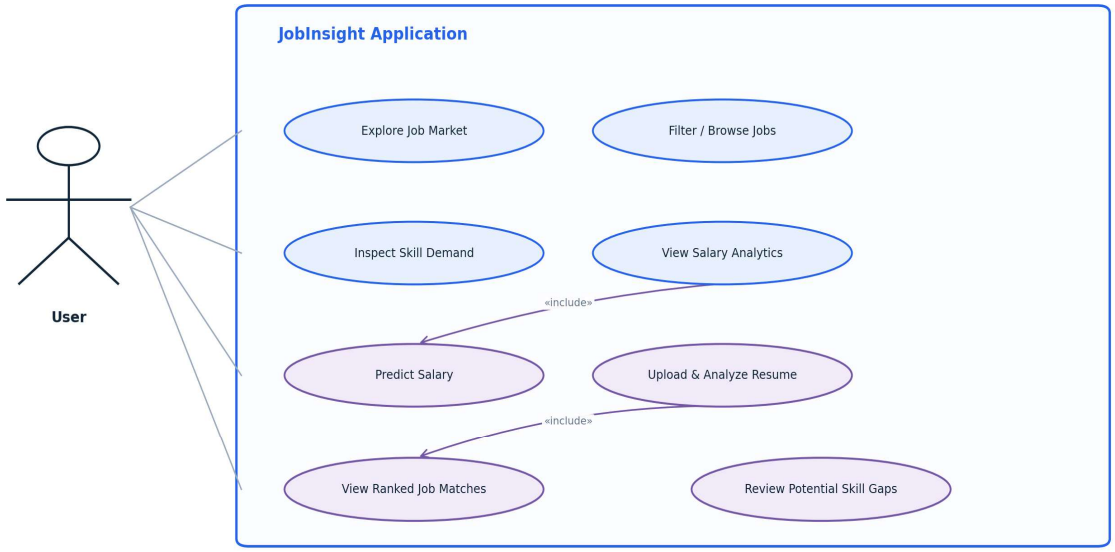


Figure 7: JobInsight Use Case Diagram

Field	Role in system	Notes
title	Job role / salary feature	Text category; normalize spelling where possible.
company_name / company	Employer information	Exact field naming depends on source dataset.
job_description	Job text	Used for skill extraction and text matching.

country / city	Location features	Location data may be incomplete or inconsistent.
salary_min / salary_max	Salary bounds	Require currency and pay-period interpretation.
years_experience_numeric	Experience feature	Numeric values may be missing or estimated.
industry / function	Job classification	Categorical model features where present.
experience_level	Seniority category	Normalize values before analysis.
education_level	Education category	Availability and consistency may vary.
work_model	Work arrangement	May represent remote, hybrid or on-site categories.
extracted_skills	Detected skill list	Derived field from the extraction module.

5.6 User Interaction Scenarios

Scenario 1 Market exploration: a job seeker opens the dashboard and reviews broad job-market summaries. The user then filters the dataset by role, location or experience level to inspect a smaller group of postings. The system should make the active filters visible so that the user understands the subset represented by the charts.

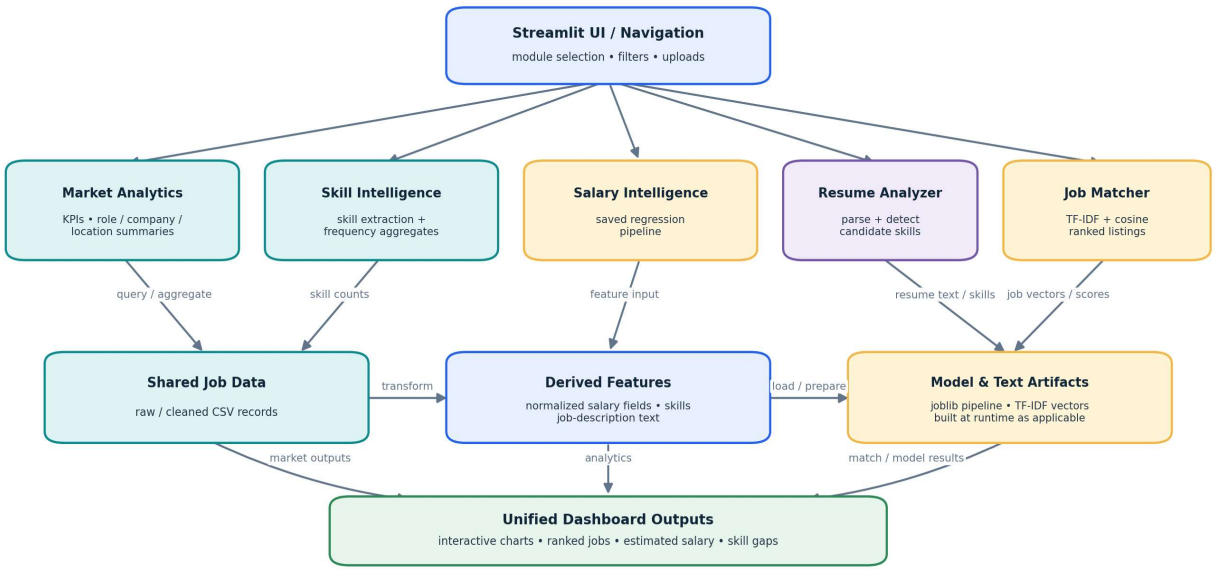


Figure 8: JobInsight Module Interaction and Data Processing Diagram

Scenario 2 Resume-based search: a user uploads a supported resume. The parser extracts text, and the matching module compares it with available job descriptions. Results are ranked by text similarity and presented with relevant listing fields. If no readable text is extracted, the system should explain the issue and invite the user to try a text-based or better-formatted file.

Scenario 3 Skill development: a user reviews the skills detected in the resume and compares them with skills frequently mentioned in selected roles. The resulting list is a starting point for further research rather than a definitive assessment. A useful interface would allow the user to select a target role and see which job descriptions contributed to the suggested skills.

Scenario 4 Salary estimation: a user supplies job attributes supported by the model. The system applies the saved preprocessing and regression pipeline to produce an estimate. The interface should show the currency and pay-period assumptions when known, and should warn when the underlying dataset does not support a reliable comparison for the selected context.

5.7 Design Principles

- Reproducibility: record data versions, model parameters, package versions and evaluation procedures.
- Privacy by design: process only necessary resume data and avoid unnecessary retention.
- Graceful degradation: empty results or unsupported files should produce understandable messages instead of unexplained crashes.
- Transparency: distinguish measured results from estimates, similarity scores and heuristic skill-gap signals.
- Consistency: the dashboard, training script and processed files should use consistent column names and category definitions.
- Separation of concerns: data preparation, analytics, model logic and presentation should remain distinct enough to test independently.

5.8 Risk Analysis

Risk	Potential impact	Mitigation
Mixed salary units	Misleading salary comparisons and model targets	Retain currency/rate metadata; standardize or separate records.
Duplicate postings	Biased analytics and optimistic test scores	Detect exact and near duplicates; consider group-aware splits.
Vocabulary gaps	Skills are missed or inconsistently counted	Maintain synonyms and manually audit a sample.
Poor resume extraction	Incorrect matches and false skill gaps	Validate extracted text; explain supported formats and limitations.
Stale derived files	Inconsistent dashboard figures	Regenerate related artifacts together and record versions.

Personal data exposure	Privacy harm and loss of user trust	Minimize retention, avoid sensitive logging and disclose handling.
Unpinned dependencies	Different behavior across machines	Pin tested versions and document environment setup.

6. EVALUATION AND TESTING

Evaluation should be carried out separately for data preparation, skill extraction, salary prediction, resume matching and the overall application. The project synopsis proposes component-specific metrics, including precision, recall and F1-score for skill extraction; MAE, RMSE and R^2 for salary prediction; ranking or classification metrics for resume-job matching; and functional, performance and usability testing for the complete system.

6.1 Salary Model Evaluation

Metric	**Meaning**	**Interpretation**
MAE	Average absolute difference between predictions and actual targets	Lower is better; target units must be known.
RMSE	Square root of mean squared prediction error	Lower is better; penalizes larger errors more heavily.
R^2	Relative fit compared with a mean-target baseline	Closer to 1 can indicate better fit, but does not alone prove generalization.

Documented figures in README: MAE = 0.415, RMSE = 0.498 and R^2 = 0.964. These figures should be verified from a fresh run and accompanied by target units, number of salary records, split details and baseline comparisons. Currency and pay-period normalization should be completed before interpreting model quality.

6.2 Skill Extraction Evaluation

A labelled sample of job descriptions should be manually annotated with expected skills. The detected skill list can then be compared with the reference labels. Precision measures the proportion of detected skills that are correct; recall measures the proportion of reference skills that were detected; F1-score balances precision and recall. No labelled evaluation results are supplied in the project materials reviewed here.

6.3 Resume Matching Evaluation

A small evaluation set can be created by having reviewers judge whether selected job listings are relevant to a resume or candidate profile. Ranking quality can be assessed using Precision@K, Recall@K or mean reciprocal rank where suitable relevance labels exist. Without such labels, cosine similarity values should be presented as similarity scores rather than accuracy measurements.

6.4 Functional Test Plan

Test ID	Test case	Expected behavior
T01	Load valid job CSV	Application reads expected columns and displays data.
T02	Dataset has missing optional fields	Application handles unavailable fields without crashing where supported.
T03	Upload valid PDF resume	Readable text is extracted where the PDF contains selectable text.
T04	Upload valid DOCX resume	Paragraph text is extracted.
T05	Upload TXT resume	Text is read and passed to matching.
T06	Upload unsupported or unreadable file	User receives a controlled error or empty-result message.

T07	Run resume matching	Ranked jobs and similarity scores are generated for available corpus.
T08	Predict salary with known categories	Pipeline returns a numeric estimate if required inputs are valid.
T09	Provide unseen categorical value	One-hot encoder handles unknown categories as configured.
T10	Review skill gaps	Potential missing skills are shown with clear caveats.

This is a proposed test plan, not evidence that every test has already been executed. Test status, actual output and defects should be filled in after running the application.

6.5 Performance and Usability

Performance testing should record application startup time, data-loading time, time to render major pages, resume-matching latency and memory usage. Usability testing should check navigation clarity, chart readability, error messages and whether users understand the meaning and limitations of similarity and salary outputs.

6.6 Evaluation Protocol and Reporting Template

For each evaluation run, record the dataset version, date of execution, code revision or archive version, package environment, random seed, train/test strategy, number of records before and after filtering, and metric values. This information makes the result interpretable and helps determine whether a change in performance is caused by the model, the data or the software environment.

A model comparison should use the same target definition and validation split for every candidate model. A simple baseline, such as predicting the median training salary, can provide context for judging whether a more complex regressor adds value. If job records are grouped by company, title or posting family, the evaluation should consider whether records from the same

group occur in both training and testing. Where the project aims to predict future postings, a time-aware split may be more realistic than a random split.

For matching evaluation, define relevance before calculating a metric. For example, reviewers could label a set of resume–job pairs as relevant, partially relevant or not relevant using agreed criteria. Precision@5 would then measure the share of the five highest-ranked results that are labelled relevant. Such an evaluation requires a documented judgement process and should not be replaced by treating the cosine similarity score itself as proof of correctness.

Evaluation record	**Information to report**
Dataset	Source, permitted use, date/version and record count.
Preprocessing	Excluded rows, duplicate handling and missing-value policy.
Salary target	Currency, rate unit, target construction and filtering rules.
Model	Algorithm, parameters, feature list and preprocessing pipeline.
Validation	Split strategy, random seed and possible group/time constraints.
Metrics	MAE, RMSE, R^2 and baseline results; include units.
Matching labels	Judgement criteria, number of resume–job pairs and reviewer procedure.
Runtime	Hardware, package versions, startup time and matching latency.

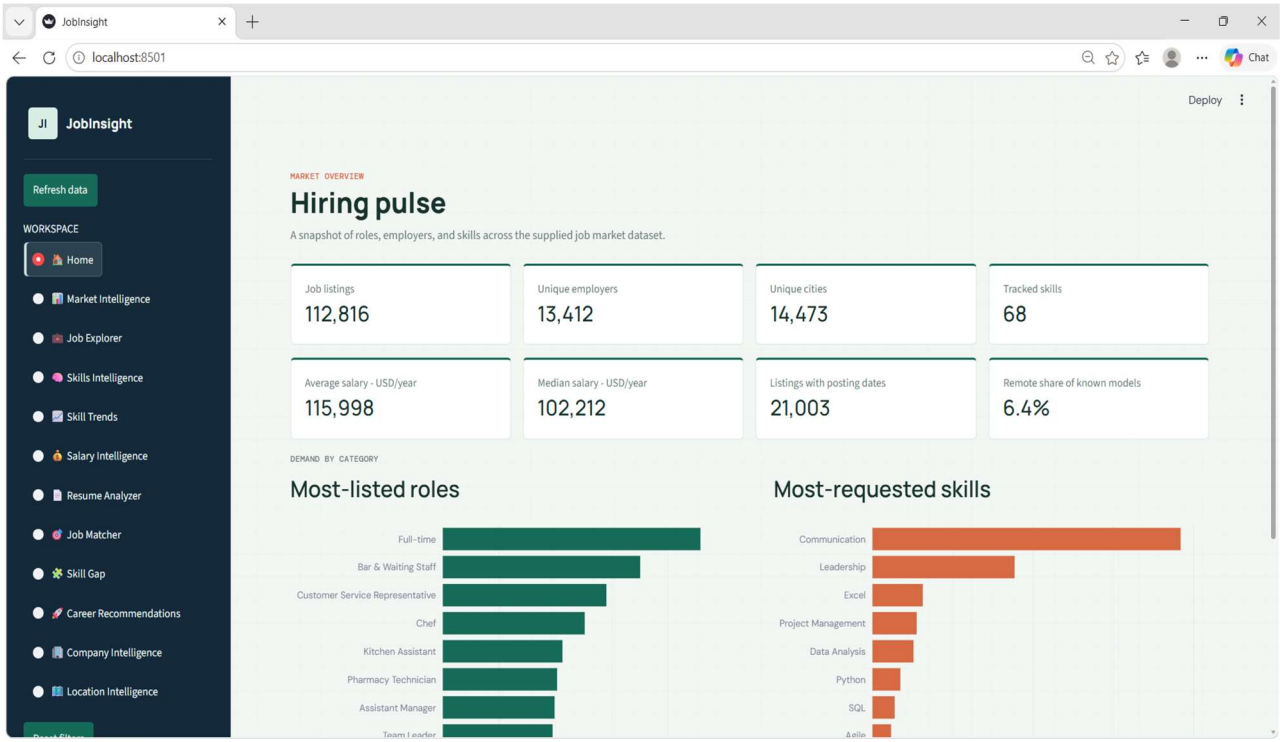
6.7 Acceptance Criteria

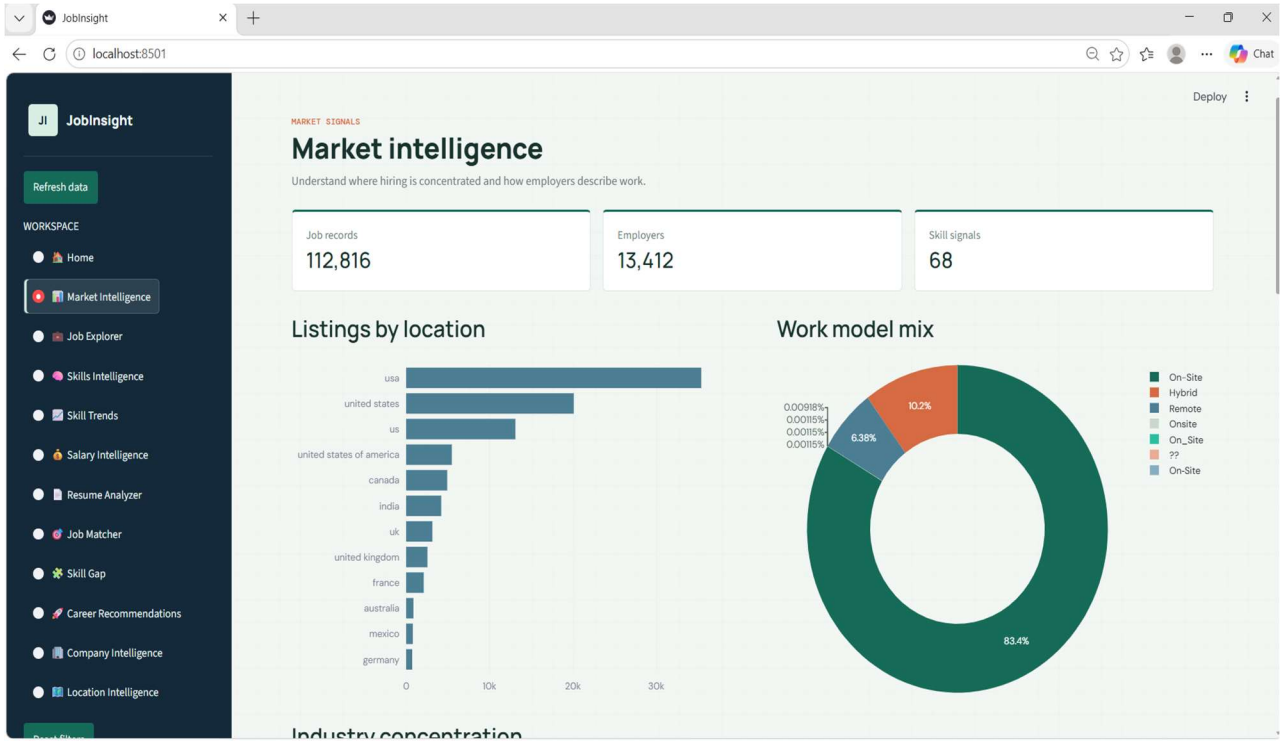
Acceptance criteria should be defined before the final demonstration. For a prototype, reasonable criteria include: the dashboard starts in the documented environment; valid input data loads; key pages render without unhandled exceptions; supported resume files return

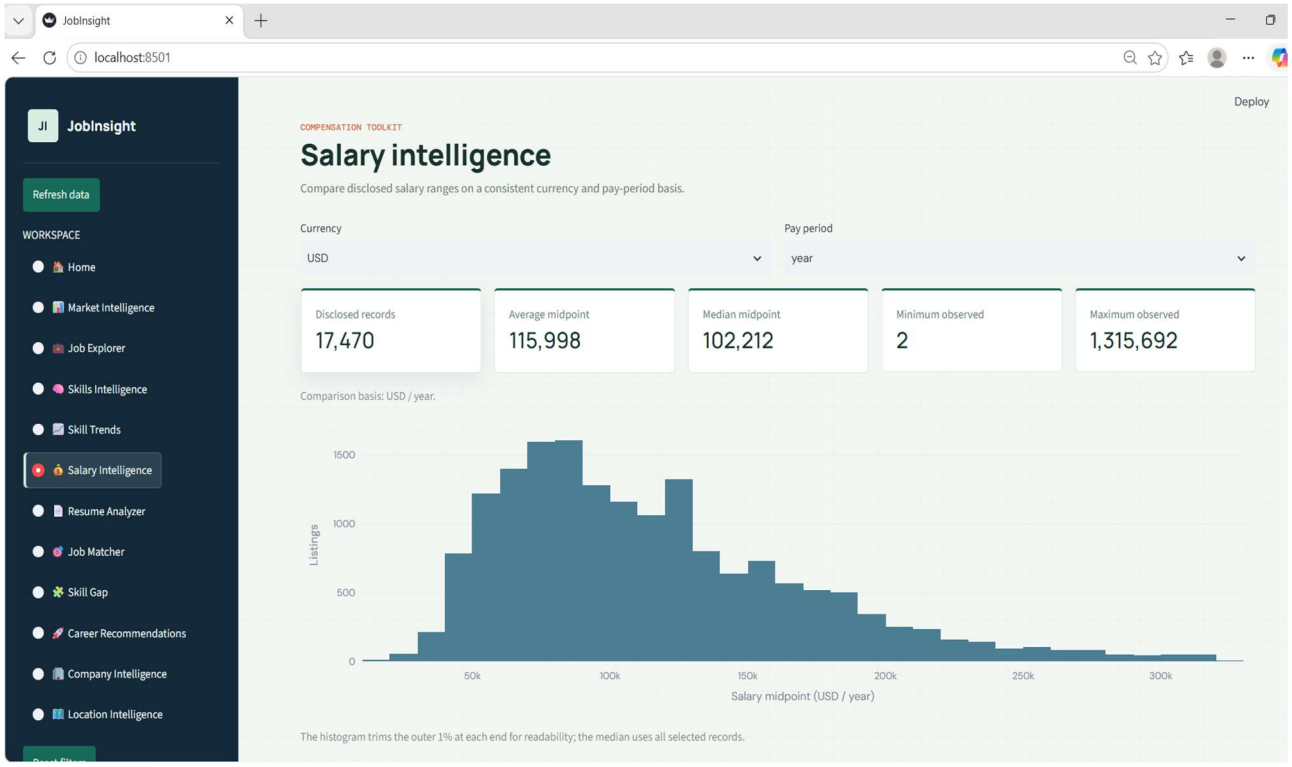
readable text where the document permits extraction; the matching pipeline returns a ranked list when eligible jobs exist; the salary pipeline returns a numeric value for valid supported inputs; and the interface explains empty results or unsupported files. Numerical model-quality thresholds should be set only after inspecting the data and establishing a baseline.

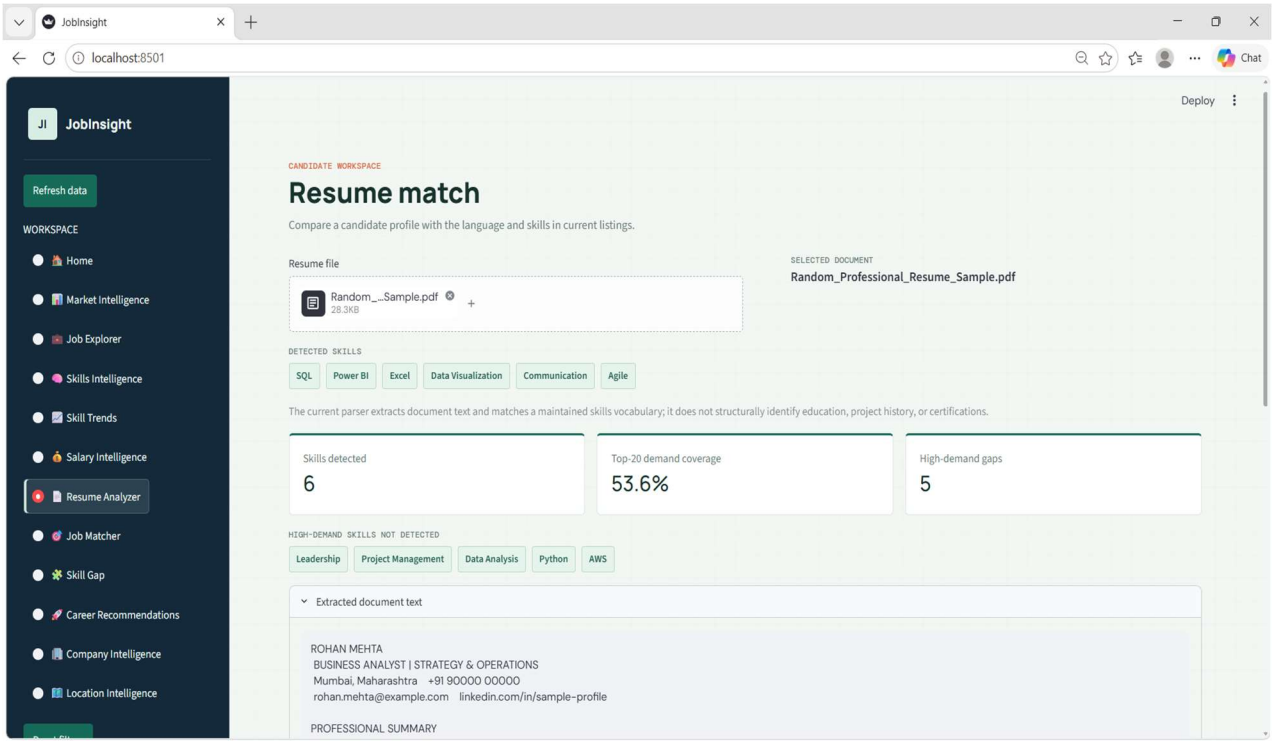
A final test report should include a status for each test case, observed output, defect identifier if applicable, and a retest result after a fix. This prevents the project report from implying that proposed tests were executed when they are only planned. It also creates a concise audit trail for academic review and future maintenance.

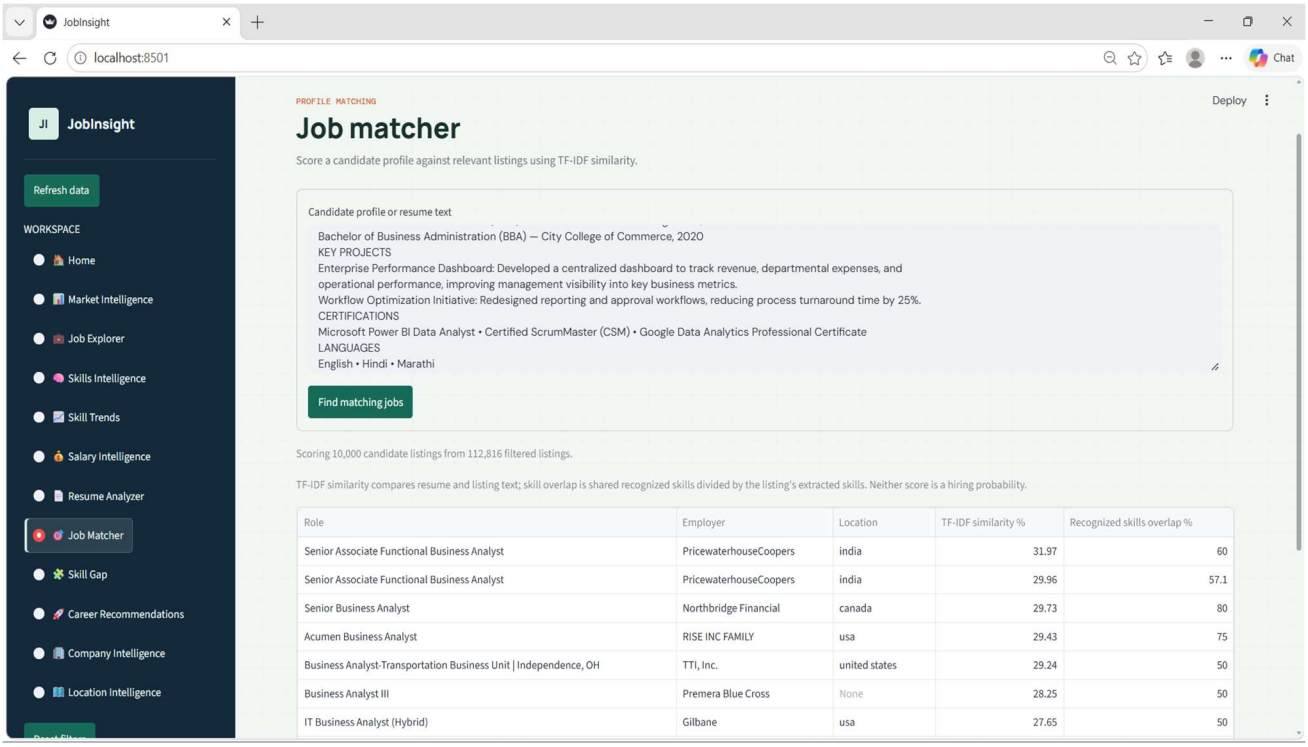
7. OUTPUT AND USER INTERFACE

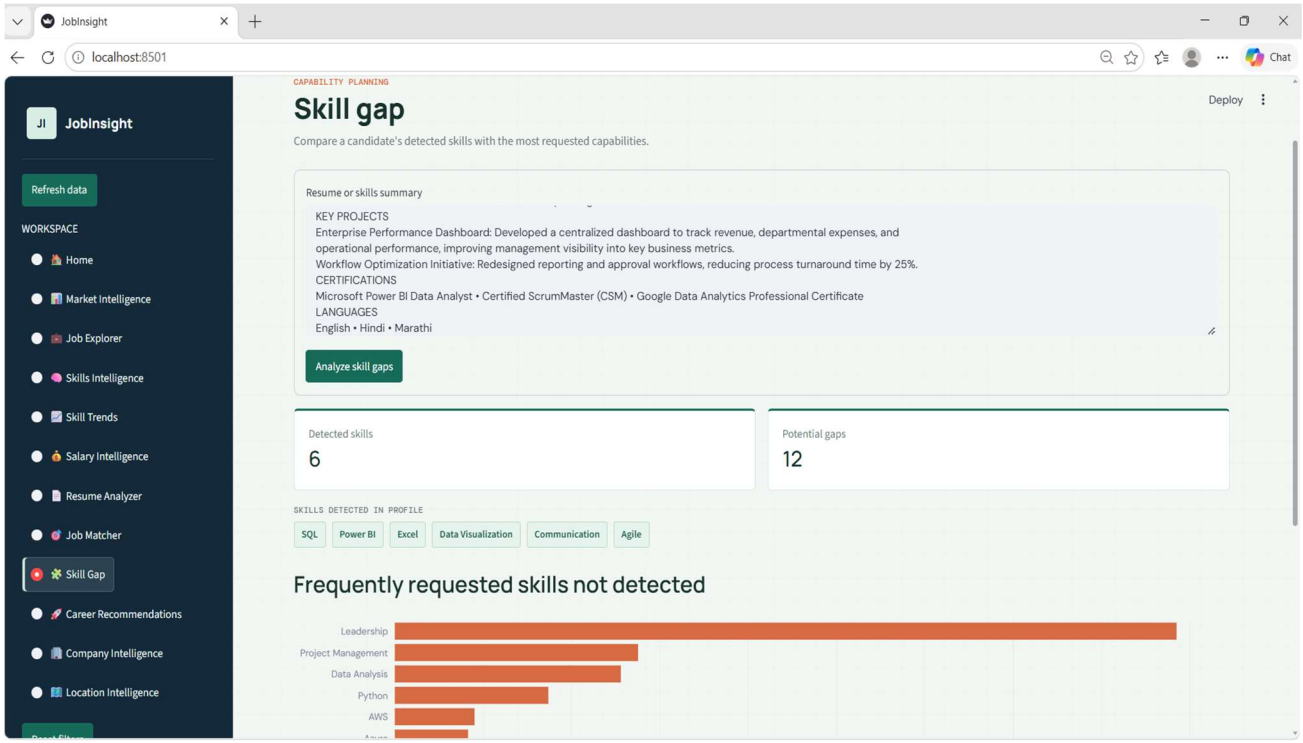



JobInsight
localhost:8501
Chat
JI
JobInsight
Deploy
Refresh data
WORKSPACE
Home
Market Intelligence
Job Explorer
Skills Intelligence
Skill Trends
Salary Intelligence
Resume Analyzer
Job Matcher
Skill Gap
Career Recommendations
Company Intelligence
Location Intelligence
MARKET SIGNALS
Market intelligence
Understand where hiring is concentrated and how employers describe work.
Job records
112,816
Employers
13,412
Skill signals
68
Listings by location
usa
united states
us
united states of america
canada
india
uk
united kingdom
france
australia
mexico
germany
0
10k
20k
30k
Work model mix
10.2%
6.38%
83.4%
On-Site
Hybrid
Remote
Onsite
On_Site
??
On-Site
Industry concentration


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Job Matcher
Skill Gap
Career Recommendations
Company Intelligence
Location Intelligence
COMPENSATION TOOLKIT
Salary intelligence
Compare disclosed salary ranges on a consistent currency and pay-period basis.
Currency
USD
Pay period
year
Disclosed records
17,470
Average midpoint
115,998
Median midpoint
102,212
Minimum observed
2
Maximum observed
1,315,692
Comparison basis: USD / year.
Listings
1500
1000
500
0
50k
100k
150k
200k
250k
300k
Salary midpoint (USD / year)
The histogram trims the outer 1% at each end for readability; the median uses all selected records.


JobInsight
localhost:8501
Chat
JobInsight
Refresh data
WORKSPACE
Home
Market Intelligence
Job Explorer
Skills Intelligence
Skill Trends
Salary Intelligence
Resume Analyzer
Job Matcher
Skill Gap
Career Recommendations
Company Intelligence
Location Intelligence
Deploy
CANDIDATE WORKSPACE
Resume match
Compare a candidate profile with the language and skills in current listings.
Resume file
Random_...Sample.pdf
28.3KB
SELECTED DOCUMENT
Random_Professional_Resume_Sample.pdf
DETECTED SKILLS
SQL
Power BI
Excel
Data Visualization
Communication
Agile
The current parser extracts document text and matches a maintained skills vocabulary; it does not structurally identify education, project history, or certifications.
Skills detected
6
Top-20 demand coverage
53.6%
High-demand gaps
5
HIGH-DEMAND SKILLS NOT DETECTED
Leadership
Project Management
Data Analysis
Python
AWS
Extracted document text
ROHAN MEHTA
BUSINESS ANALYST | STRATEGY & OPERATIONS
Mumbai, Maharashtra +91 90000 00000
rohan.mehta@example.com linkedin.com/in/sample-profile
PROFESSIONAL SUMMARY


JobInsight
localhost:8501
Chat
JobInsight
Refresh data
WORKSPACE
Home
Market Intelligence
Job Explorer
Skills Intelligence
Skill Trends
Salary Intelligence
Resume Analyzer
Job Matcher
Skill Gap
Career Recommendations
Company Intelligence
Location Intelligence
Deploy
PROFILE MATCHING
Job matcher
Score a candidate profile against relevant listings using TF-IDF similarity.
Candidate profile or resume text
Bachelor of Business Administration (BBA) — City College of Commerce, 2020
KEY PROJECTS
Enterprise Performance Dashboard: Developed a centralized dashboard to track revenue, departmental expenses, and operational performance, improving management visibility into key business metrics.
Workflow Optimization Initiative: Redesigned reporting and approval workflows, reducing process turnaround time by 25%.
CERTIFICATIONS
Microsoft Power BI Data Analyst • Certified ScrumMaster (CSM) • Google Data Analytics Professional Certificate
LANGUAGES
English • Hindi • Marathi
Find matching jobs
Scoring 10,000 candidate listings from 112,816 filtered listings.
TF-IDF similarity compares resume and listing text; skill overlap is shared recognized skills divided by the listing's extracted skills. Neither score is a hiring probability.
Role
Employer
Location
TF-IDF similarity %
Recognized skills overlap %
Senior Associate Functional Business Analyst
PricewaterhouseCoopers
india
31.97
60
Senior Associate Functional Business Analyst
PricewaterhouseCoopers
india
29.96
57.1
Senior Business Analyst
Northbridge Financial
canada
29.73
80
Acumen Business Analyst
RISE INC FAMILY
usa
29.43
75
Business Analyst-Transportation Business Unit | Independence, OH
TTI, Inc.
united states
29.24
50
Business Analyst III
Premera Blue Cross
None
28.25
50
IT Business Analyst (Hybrid)
Gilbane
usa
27.65
50



8. LIMITATIONS AND DRAWBACKS

- Vocabulary-based skill extraction cannot reliably recognize every synonym, new skill or contextual meaning.
- TF-IDF matching is based on lexical patterns and may rank textually similar jobs even when their actual requirements differ.
- Salary fields may be incomplete and may mix currencies or pay periods; the model should not be treated as a universal salary estimator until these are normalized.
- The documented salary metrics have not been independently reproduced as part of this report preparation.
- A random train/test split may overstate generalization if duplicates or closely related job postings occur in both partitions.
- Resume parsing may fail on scanned PDFs, complex layouts, tables, images or unusual formatting.
- Potential skill gaps may arise because the resume does not mention a skill, not because the candidate lacks it.

- CSV-based processing and full text vectorization can become resource-intensive as the dataset grows.
- The supplied requirements file does not pin package versions, which can reduce reproducibility across environments.
- The reviewed repository includes a local data-inspection script referencing a machine-specific file path; this is not portable without adjustment.
- Job-market insights reflect the supplied dataset and its collection period; they may not represent the entire market or current openings.
- Resume information can contain personal data and should be handled with appropriate consent, access controls and retention practices.

9. FUTURE ENHANCEMENTS

- Expand skill vocabulary and normalize aliases, abbreviations and related terms.
- Evaluate sentence-transformer embeddings for semantic resume-job matching against the current TF-IDF baseline.
- Build a labelled skill-extraction benchmark and compare rule-based methods with supervised or transformer-based models.
- Standardize salary values to a common currency and pay period when reliable conversion information is available.
- Compare Random Forest with baseline regression, gradient boosting and other suitable models using consistent evaluation splits.
- Use group-aware or time-aware validation to reduce leakage from repeated or similar postings.
- Add explainable match evidence, including overlapping skills, relevant experience and unmet listed requirements.
- Add OCR support for scanned resumes only if it can be implemented and tested responsibly.
- Move from local CSV files to a database if concurrent users and larger datasets require it.
- Pin dependency versions and create a repeatable setup or deployment configuration.

- Add automated tests for preprocessing, parsing, matching, prediction and dashboard input validation.
- Improve resume privacy through temporary processing, minimal retention and clear user notices.
- Add data freshness indicators and source metadata so users can judge how current the market analysis is.

10. CONCLUSION

JobInsight is designed as an integrated job-market analytics and career recommendation platform. It combines descriptive analysis of job postings with skill extraction, salary estimation, resume parsing, text-based job matching and potential skill-gap identification. The system is implemented as a Python and Streamlit prototype supported by CSV datasets and a saved machine-learning pipeline.

The project's central contribution is the integration of market-level information with candidate-oriented exploration in one dashboard. The chosen baseline methods are comparatively interpretable and practical for a prototype. At the same time, the quality of the outputs depends on dataset representativeness, skill-vocabulary coverage, resume text extraction and salary data consistency.

The next stage of development should focus on reproducible evaluation, salary normalization, matching-quality assessment, automated tests and clearer explanations of recommendation scores. With these improvements, JobInsight can provide a more reliable and transparent tool for exploring job requirements and planning skills development.

Project Outcome Summary

Area	Outcome described by project	Verification status in this report
Job-market analytics	Dashboard summaries and visual exploration	Feature documented; screenshots should be added from running app.
Skill extraction	Predefined vocabulary-based extraction	Implementation described from supplied code.
Salary prediction	Random Forest regression pipeline	Implementation described; reported metrics need reproduction.
Resume parsing	PDF/DOCX/TXT text extraction	Supported formats documented; edge cases need testing.
Resume matching	TF-IDF and cosine similarity	Implementation described; ranking quality needs labelled evaluation.
Skill-gap analysis	Compare resume skill signals with market/job skills	Interpret as potential gaps, not definitive skill deficiencies.

11.1 Practical Significance

The practical value of JobInsight is its attempt to reduce the effort required to move between broad market research and individual job exploration. A candidate can review the kinds of skills mentioned in a dataset, inspect role and location patterns, and compare a resume with available descriptions in the same interface. This can support planning and learning, provided that users understand that the data may be incomplete and the algorithms are not substitutes for human judgement.

11.2 Recommendations for Responsible Use

- Re-evaluate the model when the dataset, feature definitions or preprocessing rules change.
- Keep resume data private and avoid storing uploaded files or extracted text unnecessarily.

- Describe unrecognized skills as “not detected” rather than asserting that the candidate lacks them.
- Show salary currency and rate unit wherever they are known, and avoid unsupported cross-market comparisons.
- Do not treat similarity scores as hiring probabilities or guaranteed fit scores.
- Explain the data source, collection period and limitations of the job dataset.
- Use the dashboard as a decision-support tool, not as a hiring or rejection system.

12. REFERENCES / BIBLIOGRAPHY

[1] Lothe, D. M., Tiwari, P., Patil, N., Patil, S., & Patil, V. (2021). “Salary Prediction Using Machine Learning.” International Journal of Advance Scientific Research and Engineering Trends, 6(5), May 2021.

[2] Quan, T. Z., & Raheem, M. (2022). “Salary Prediction in Data Science Field Using Specialized Skills and Job Benefits – A Literature Review.” Journal of Applied Technology and Innovation, 6(3), 2022. https://jati.apu.edu.my/index.php/JATI/article/view/181

[3] Mamidala, J. V., et al. “Machine Learning Approaches to Salary Prediction in Human Resource Payroll Systems.” Journal of Computer Science and Technology Studies. Available records conflict on year/volume/issue details. DOI: https://doi.org/10.32996/jcsts.2025.7.10.52.

[4] Almgerbi, M., De Mauro, A., Kahlawi, A., & Poggioni, V. (2022). “A Systematic Review of Data Analytics Job Requirements and Online-Courses.” Journal of Computer Information Systems, 62(2), 422–434. Published online in 2021. DOI: https://doi.org/10.1080/08874417.2021.1971579

[5] Alibasic, A., Upadhyay, H., Simsekler, M. C. E., Kurfess, T., Woon, W. L., & Omar, M. A. (2022). "Evaluation of the Trends in Jobs and Skill-Sets Using Data Analytics: A Case Study." Journal of Big Data, 9, Article 32. DOI: https://doi.org/10.1186/s40537-022-00576-5

[6] Lukauskas, M., Šarkauskaitė, V., Pilinkienė, V., Stundžienė, A., Grybauskas, A., & Bruneckienė, J. (2023). "Enhancing Skills Demand Understanding through Job Ad Segmentation Using NLP and Clustering Techniques." Applied Sciences, 13(10), Article 6119. DOI: https://doi.org/10.3390/app13106119

[7] Bhola, A., Halder, K., Prasad, A., & Kan, M.-Y. (2020). "Retrieving Skills from Job Descriptions: A Language Model Based Extreme Multi-label Classification Framework." Proceedings of COLING 2020, 5832–5842. DOI: https://doi.org/10.18653/v1/2020.coling-main.513

[8] Nguyen, K., Zhang, M., Montariol, S., & Bosselut, A. (2024). "Rethinking Skill Extraction in the Job Market Domain using Large Language Models." Proceedings of NLP4HR 2024, 27–42. DOI: https://doi.org/10.18653/v1/2024.nlp4hr-1.3

[9] Senger, E., Zhang, M., van der Goot, R., & Plank, B. (2024). "Deep Learning-based Computational Job Market Analysis: A Survey on Skill Extraction and Classification from Job Postings." Proceedings of NLP4HR 2024, 1–15. DOI: https://doi.org/10.18653/v1/2024.nlp4hr-1.1

[10] Yan, R., Le, R., Song, Y., Zhang, T., Zhang, X., & Zhao, D. (2019). "Interview Choice Reveals Your Preference on the Market: To Improve Job-Resume Matching through Profiling Memories." Proceedings of KDD '19, 914–922. DOI: https://doi.org/10.1145/3292500.3330963

[11] Dave, V. S., Zhang, B., Hasan, M., Aljadda, K., & Korayem, M. (2018). "A Combined Representation Learning Approach for Better Job and Skill Recommendation." Proceedings of CIKM '18. DOI: https://doi.org/10.1145/3269206.3272023

[12] Gurcan, F., et al. (2025). Two potentially relevant publications were identified; confirm which one matches the synopsis: (a) “Towards a Sustainable Workforce in Big Data Analytics: Skill Requirements Analysis from Online Job Postings Using Neural Topic Modeling.” Sustainability, 17(20), Article 9293. DOI: https://doi.org/10.3390/su17209293; or (b) “Future-Ready Skills Across Big Data Ecosystems: Insights from Machine Learning-Driven Human Resource Analytics.” Applied Sciences, 15(11), Article 5841. DOI: https://doi.org/10.3390/app15115841.

[13] Shahzada, A., Hussain, I., Shaheen, N., Hussain, S. M., Khan, T. F., & Shehzad, M. (2025). “CAREERLLAMA: An AI-Powered Personalized Career Recommendation System with Psychometric and Skill Gap Integration.” Spectrum of Engineering Sciences, 3(6), 404–414. DOI: https://doi.org/10.5281/zenodo.15674001